



ELiteTEch Intern

Learn -- Innovate -- Thrive

Internship Program



Machine Learning



TASK INSTRUCTION

- MUST ATTEMPT SIX OUT OF FOUR TASKS
- SUBMIT BEFORE DEADLINE
- USE INTERNET RESOURCES
- FOLLOW GROUP UPDATES

INSTRUCTION

- STORE ALL YOUR WORK-RELATED CODE AND FILES IN A GITHUB REPOSITORY.
- MAINTAIN PROPER COMMENTING IN YOUR CODE FOR BETTER UNDERSTANDING AND READABILITY.
- USE RESOURCES LIKE YOUTUBE, GOOGLE, AND CHATGPT TO HELP YOU COMPLETE TASKS EFFICIENTLY.
- FOLLOW THE VIDEO GUIDANCE PROVIDED IN WHATSAPP WHILE SUBMITTING YOUR TASKS.
- COMPLETE ALL FOUR TASKS AND SUBMIT THEM WITHIN THE DEADLINE WHILE WORKING AT YOUR CONVENIENCE.

TASK

1



DECISION TREE IMPLEMENTATION

- BUILD AND VISUALIZE A DECISION TREE MODEL USING SCIKIT-LEARN TO CLASSIFY OR PREDICT OUTCOMES ON A CHOSEN DATASET
- DELIVERABLE: A NOTEBOOK WITH MODEL VISUALIZATION AND ANALYSIS.

**COMPLETION CERTIFICATE WILL BE
ISSUED AFTER THE END OF THE
INTERNSHIP**



TASK

2



SENTIMENT ANALYSIS WITH NLP

- PERFORM SENTIMENT ANALYSIS ON A DATASET OF CUSTOMER REVIEWS USING TF-IDF VECTORIZATION AND LOGISTIC REGRESSION.
- DELIVERABLE: A JUPYTER NOTEBOOK SHOWCASING PREPROCESSING, MODELING, AND SENTIMENT EVALUATION

**COMPLETION CERTIFICATE WILL BE
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INTERNSHIP**



TASK

3

IMAGE CLASSIFICATION MODEL

- BUILD A CONVOLUTIONAL NEURAL NETWORK (CNN) FOR IMAGE CLASSIFICATION USING TENSORFLOW OR PYTORCH.
- DELIVERABLE: A FUNCTIONAL MODEL WITH PERFORMANCE EVALUATION ON A TEST DATASET.



**COMPLETION CERTIFICATE WILL BE
ISSUED AFTER THE END OF THE
INTERNSHIP**



TASK

4



RECOMMENDATION SYSTEM

- BUILD A RECOMMENDATION SYSTEM USING COLLABORATIVE FILTERING OR MATRIX FACTORIZATION TECHNIQUES.
- DELIVERABLE: A NOTEBOOK OR APP SHOWCASING RECOMMENDATION RESULTS AND EVALUATION METRICS.

COMPLETION CERTIFICATE WILL BE
ISSUED AFTER THE END OF THE
INTERNSHIP



ASK US FOR HELP!



**THE PURPOSE OF
THIS INTERNSHIP
IS
TO LEARN AND
GROW**

PERSONAL DEVELOPMENT

**THIS INTERNSHIP
WILL HELP YOU TO
ENHANCE YOUR
EMPLOYABILITY
PROSPECTS AND
RAISE YOUR
CONFIDENCE.**

**WE DON'T WANT
TO DICTATE
YOU. IT IS UP TO
YOU TO SEEK
GUIDANCE.**

**THE TASKS GIVEN
MAY SEEM VERY
EASY OR VERY
DIFFICULT. WE
EXPECT THAT YOU
GIVE PROFESSIONAL
DUE REGARD TO THE
TASKS.**

Thank
You

The image features a background with a horizontal gradient from blue on the left to pink on the right. A large, stylized, multi-colored letter 'S' is positioned behind the text. The 'S' is composed of several overlapping, semi-transparent shapes in shades of pink, light blue, and light purple. The text 'Thank You' is written in a black, cursive script font, with 'Thank' on the top line and 'You' on the bottom line, centered horizontally.